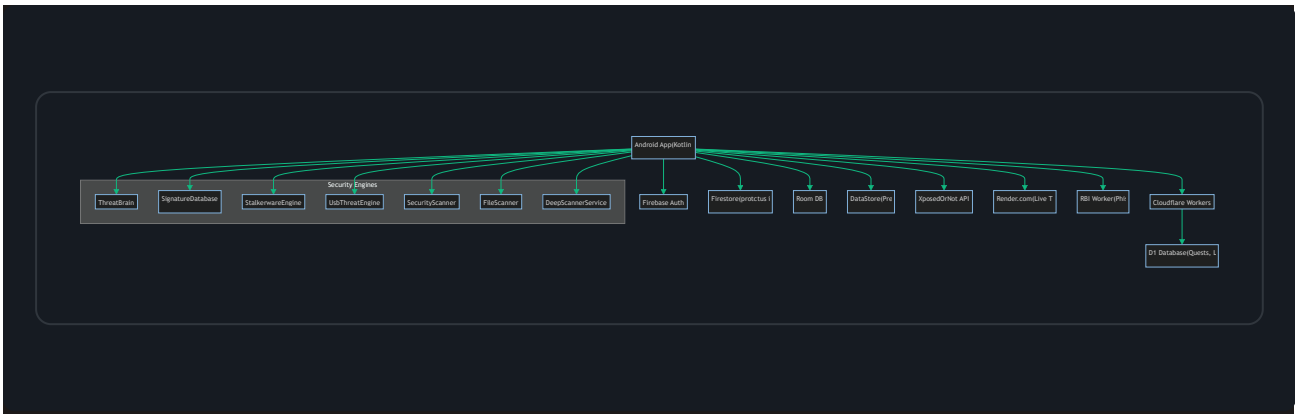
Protctus — Complete Application Analysis Report

Date: April 4, 2026
Codebase: Android (Kotlin / Jetpack Compose), Cloudflare Workers (TypeScript), Firebase
Architecture: MVVM + Hilt DI + Room + DataStore + Coroutines

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1. Architecture Overview



Layer	Technology	Purpose
UI	Jetpack Compose + Material 3	All screens
DI	Hilt	16 ViewModels injected
State	StateFlow + MutableStateFlow	Reactive UI
Local DB	Room (v7, 8 entities)	Threats, Settings, VVIP, SafeList
Preferences	DataStore	11 toggleable settings
Auth	Firebase Auth + Email OTP	Sign-up, Sign-in, 2FA, OTP Security

2. Complete Feature Inventory

SECURITY ENGINES (7 Real Engines)

#	Feature	Status	How It Works
1	**ThreatBrain Heuristic Engine**	Real	Multi-factor behavioral analysis: permission combos, headless detection.

2	**Bloom Filter Signature DB**	Real	MurmurHash3 Bloom Filter with 64KB BitSet, 3 hash functions.
3	**Stalkerware Detection**	Real	4-rule scoring: sideloading, accessibility/admin, hidden icon.
4	**USB Juice Jacking Guard**	Real	Real-time USB state monitoring. SOS Morse vibration.
5	**Deep Scanner Service**	Real	Foreground Service with WakeLock for recursive system scanning.
6	**File Scanner**	Real	Concurrent coroutine-batched scanner (8 files/chunk).
7	**Device Security Scanner**	Real	12 real checks across 3 categories with scored findings.

3. Security Engine Deep-Dive — "World's Fastest"

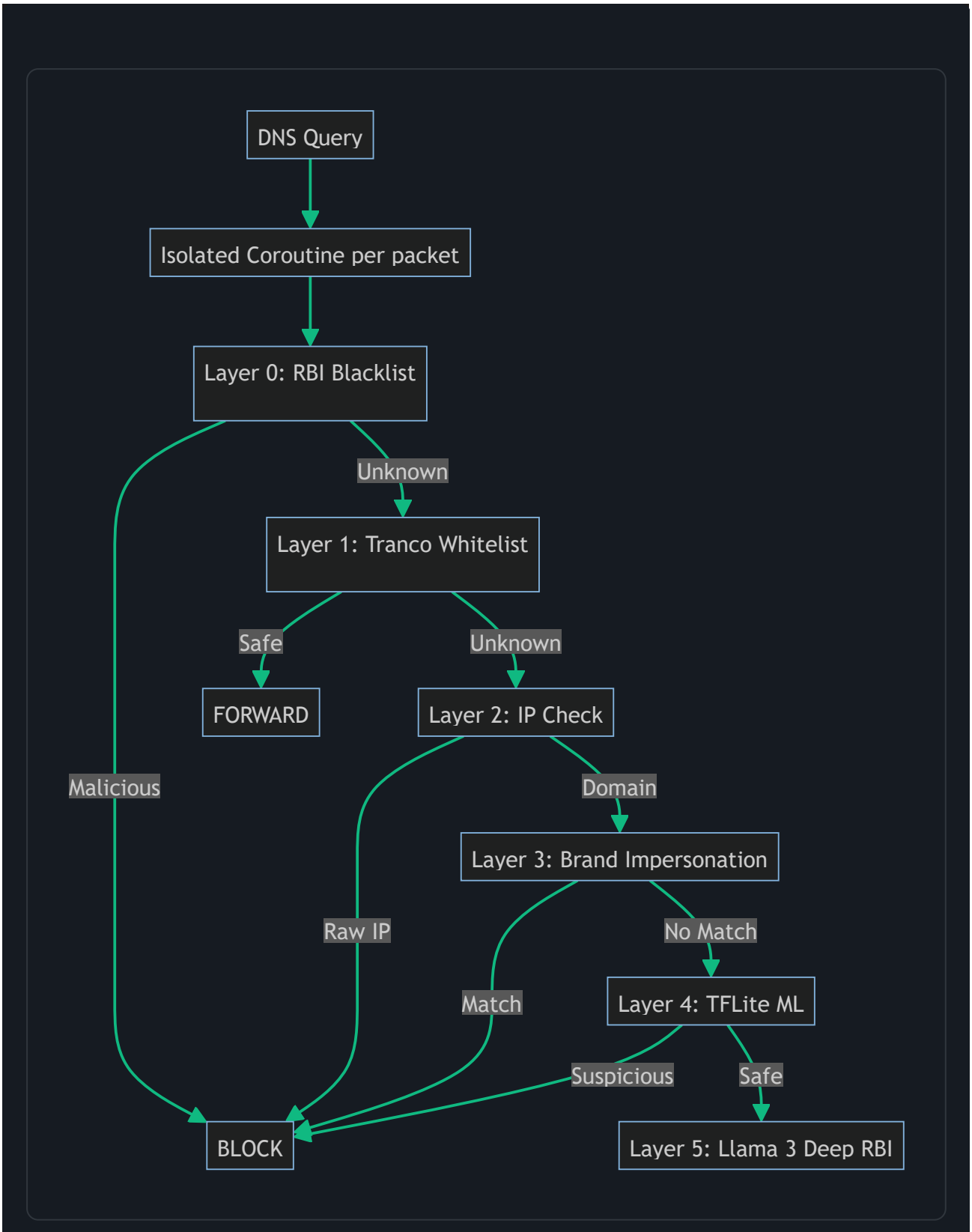
> [!IMPORTANT] The engine's speed comes from its **architectural design** — it never downloads definitions at runtime and uses probabilistic data structures.

Speed Factor: Bloom Filter (O(1) Lookup)

Protctus uses a **64KB Bloom Filter** with MurmurHash3.

- **3 hash probes per lookup** — each takes ~100ns.
- **Total lookup:** ~300ns per file vs. ~50ms for a linear DB scan.

4. Phishing Engine — 5-Layer AI DNS Filter



5. Backend Infrastructure

Worker	Purpose
Quests API	Admin panel, Quests, Lab, Ticket management.
Leaderboard	Xp tracking and global user ranking.
RBI Phishing	Deep browser-based phishing analysis.
OTP Auth	Secure Email OTP generation and verification.

6. Data Layer Architecture

Room Database (v7, 8 Entities)

Entity	Use
`ThreatAlert`	All security findings and scan history.
`UserSafeList`	User-whitelisted domains.
`VvipDomain`	Tranco Top-380K trusted domains.
`OtpState`	Advanced rate limiting and cooldown logic.

7. UI/UX Architecture

The app contains **47+ screens** built with Jetpack Compose.

Category	Screens
Dashboard	Home, RankDefend, Rewards, Live Threat Map.
Scanner	SmartScan, DeepScan, URL Scanner, Dark Web Scan.

Lab	Chapters, Topics, Microlearning, Quizzes.	

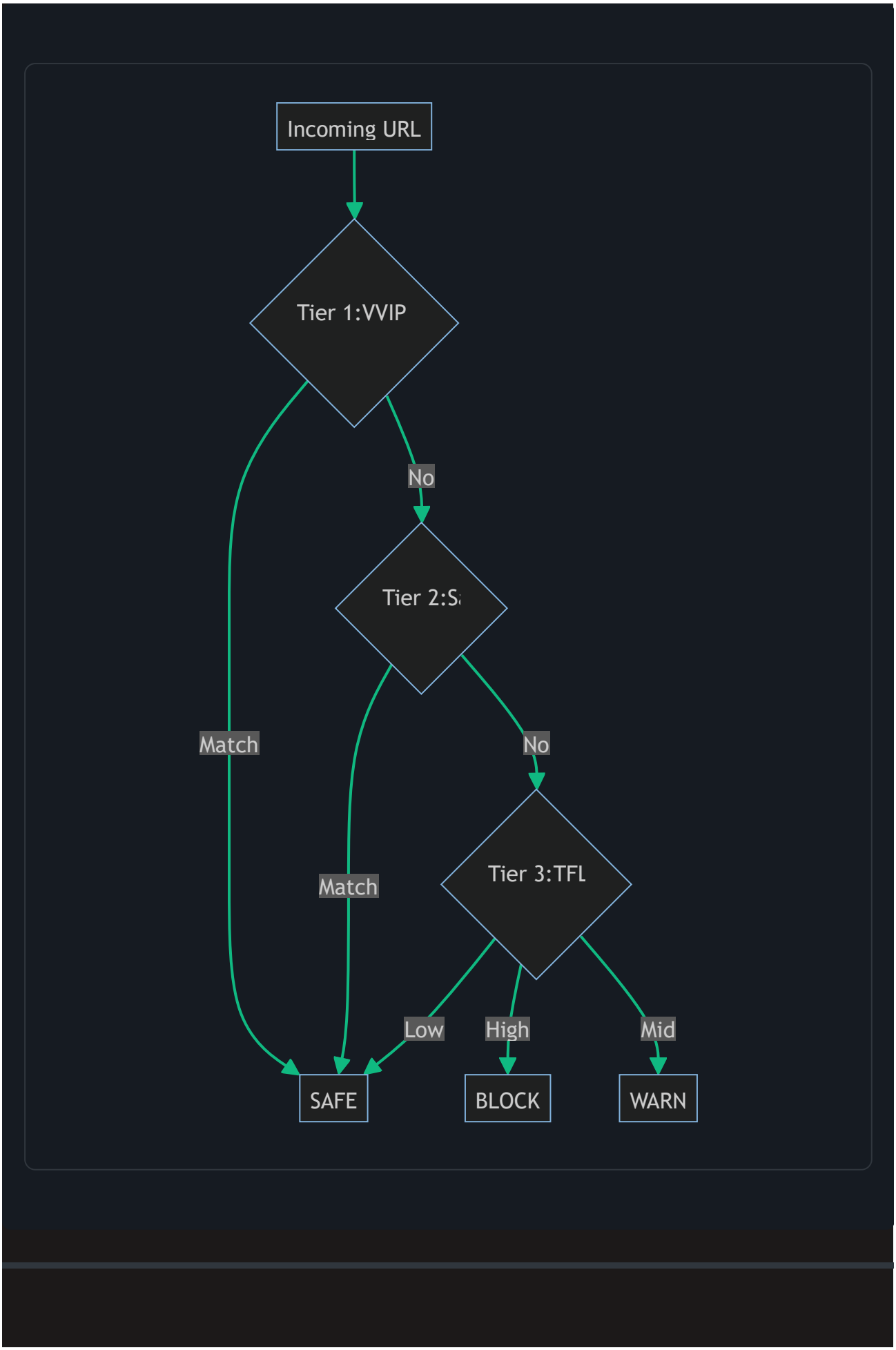
8. Gamification System

Level	Rank	XP Required
1	Cyber Novice	0
4	Virus Slayer	600
8	Cyber Knight	4000

9. App Self-Security

- **Certificate Pinning**: Cloudflare ECC CA-3 + DigiCert P384.
- **Hardware Keystore**: AES-256-GCM encryption.
- **Native Anti-Tamper**: Chakravyuh JNI module.

10. 3-Tier Phishing Defense Architecture


Incoming URL
Tier 1:VVIP
No
Tier 2:S
Match
No
Match
Tier 3:TFL
Low
High
Mid
SAFE
BLOCK
WARN

11. Device Requirements

Requirement	Specification
OS	Android 8.0+ (Oreo)
RAM	3GB Minimum (4GB Recommended)
Storage	~65MB Install / ~150MB Runtime

12. APK Build Audit

